Some basic instructions:

- If the question says, "**Write a function**" or "**Write a user-defined function**", then **DO NOT** write any other code (i.e. do not write main function, do not read input, do not write code for calling the functions). Writing those codes is not wrong but does not carry any marks. It will be a waste of time.
- If the question says, "**Write a program**" or "**Write a C program**", then you have to write the complete code, including main function, input reading etc.
- Practice at least one sorting (bubble sort may not be the most efficient but will be easier to implement).
- Do not explain the output or code, if it is not asked.

1. Write a function that receives name of 100 people (where each name is at most 100 characters long). The function prints the valid names only. A name is considered as invalid, if it contains any other characters except lowercase and uppercase alphabets.
2. Given an array of strings, print the strings sorted in the ascending order based on their length.

   Input Strings:
   helloABC
   World
   abc123

   Output Strings:
   World
   abc123
   helloABC

3. Given an array of strings, find the sum of ASCII values of all the lower-case letters for each string.

   Input Strings:
   helloABC
   World
   abc123

   Output Strings:
   helloABC = 532
   World = 433
   abc123 = 294

4. Given a 2D integer array, count how many odd numbers are present in each row

   Input Array:

1 2 3
2 2 2
3 3 3

Output:
Row 1: 1
Row 2: 0
Row 3: 3

5. Given the values of the pixels of an image as a 2D integer array, change the values of each pixel by taking the average of all surrounding pixels.
6. Write a function simulating the following library functions:
    a. strlen
    b. strcat
    c. strstr
7. Write a recursive function that receives a string, and checks whether it is palindrome or not
8. Write a recursive function that receives a string and prints its reverse
9. Write a recursive function that receives an integer and returns the sum of its digits
10. Write a recursive function to simulate the problem tower of Hanoi
11. Declare the structure to store the following information of a person:
    a. Name
    b. Age
    c. Address (it is a structure)

    The Address structure will have this information:
    a. Road number
    b. House number
    c. District
    d. Post Code

        Write a function that, receives an array of person structure, and sort them based on post code in ascending order
12. Given an array of person structures containing information about 10 peoples, write a function to find how many people live in a specific district.
13. Write a function that receives an address of a person structure and reads all the information from the user and saves into the structure.

14. Write a function that receives two 2D integer array and their dimensions (dimension will be same for both array). The function should create a new array by picking the largest value between the same locations of both arrays. The function will return a reference to the new array.

    Input Arrays:
    1 2 5
    2 8 9
    1 1 6

    4 4 4
    3 3 3
    5 5 5

    Output Array:
    4 4 5
    3 8 9
    5 5 6

15. Write a program that reads a C program from a folder, and creates 5 copies of it
16. Write a program that creates a copy of its own source code (explore about predefined macro named __FILE__)
17. Write a program that can save information about 100 persons (previously declared structure) into a file
18. Write a program that can load information about 100 persons (previously declared structure) from a file
19. Write a program, that reads 4 integer values separated by – symbol from each line of a text file and prints the sum of the numbers for each line.

    Input file:
    1-2-3-3
    2-2-2-2
    3-3-3-3

    Output:
    9
    8
    12